

Cluster algorithms

- *K – means*
- *Heirarchical*

Can I assume ,If i give any data set will you able to develop ML classification model

- 1) *Read the data (Git hub,SQL server)*
- 2) *EDA or preprocess (Statistics)*
- 3) *Model development*
- 4) *Model evaluation*
- 5) *Hyper parameter tuning*
- 6) *Model deployment*

ML divided supervised and unsupervised

Supervised : we have a target column.we have a labeled data

Unsupervised: we dont have a target column

we have data only

There is no end goal

we divide data into some groups

- *segment the data*
- *Cluster the data*

we have mainly two algorithms

- *K – means*
- *Heirarchical cluster*

K – means

***here k is hyper parameter , it is a number
mean = average***

Step – 1: All the data points divided randomly into groups

Step – 2: Will calculate mean of every group

*Step – 3: we will calculate distance between each point of all the groups
with all the group means*

Step – 4: Based on distances the rearrange of groups will form

Step – 5: Repeat the process step – 1,2,3,4 multiple times

Till the group means are not changes

Step – 6: Then will get our final clusters

$k = 2$ means in starting we are dividing into two groups

$k = 3$ we are choosing 3 groups

$k =$ number of clusters or groups

will calculate distance

will use Euclidean distance

- 1) Euclidean distance*
- 2) Cosine distance*
- 3) Manhattan distance*
- 4) Jacobian distance or jaccard distance*

Individual	Variable 1	Variable 2
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

$$\text{Distance: } (x_1, y_1) (x_2, y_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Step – 1: Randomly divide the groups based k value

$$K = 2 \text{ (Two groups)}$$

Group – 1 : (1,2,3)observations

$$(1,1) (1.5,2) (3,4)$$

Group – 2 : (4,5,6,7)observations

$$(5,7) (3.5,5) (4.5,5), (3.5,4.5)$$

$$\text{Mean } (G_1) = \frac{(1 + 1.5 + 3)}{3}, \frac{1 + 2 + 4}{3} = (1.83, 2.33)$$

$$\text{Mean } (G_2) = \frac{(5 + 3.5 + 4.5 + 3.5)}{4}, \frac{7 + 5 + 5 + 4.5}{4} = (4.125, 5.375)$$

Step – 2: Calculate distance between each point with both group means (centroids)

Group – 1 points with Group – 1 mean

$$(1,1) \text{ vs } (1.83, 2.33): \sqrt{(1.833 - 1)^2 + (2.33 - 1)^2} = 1.57$$

$$(1.5, 2) \text{ vs } (1.83, 2.33): \sqrt{(1.833 - 1.5)^2 + (2.33 - 2)^2} = 0.47$$

$$(3, 4) \text{ vs } (1.83, 2.33): \sqrt{(1.833 - 3)^2 + (2.33 - 4)^2} = 2.04$$

Group – 1 points with Group – 2 mean

$$(1,1) \text{ vs } (4.125, 5.5): \sqrt{(4.125 - 1)^2 + (5.5 - 1)^2} = 7.21$$

$$(1.5, 2) \text{ vs } (4.125, 5.5): \sqrt{(4.125 - 1.5)^2 + (5.5 - 2)^2} = 6.10$$

$$(3, 4) \text{ vs } (4.125, 5.5): \sqrt{(4.125 - 3)^2 + (5.5 - 4)^2} = 1.78$$

Group – 2 points with Group – 1 mean

$$(5, 7) (3.5, 5) (4.5, 5), (3.5, 4.5)$$

$$(5, 7) \text{ vs } (1.83, 2.33): \sqrt{(1.833 - 5)^2 + (2.33 - 7)^2} = 5.64$$

$$(3.5, 5) \text{ vs } (1.83, 2.33): \sqrt{(1.833 - 3.5)^2 + (2.33 - 5)^2} = 3.15$$

$$(4.5, 5) \text{ vs } (1.83, 2.33): \sqrt{(1.833 - 4.5)^2 + (2.33 - 5)^2} = 3.78$$

$$(3.5, 4.5) \text{ vs } (1.83, 2.33): \sqrt{(1.833 - 3.5)^2 + (2.33 - 4.5)^2} = 2.74$$

Group – 2 points with Group – 2 mean

$$(5,7) \text{ vs } (4.125,5.5): \sqrt{(4.125 - 5)^2 + (5.5 - 7)^2} = 1.84$$

$$(3.5,5) \text{ vs } (4.125,5.5): \sqrt{(4.125 - 3.5)^2 + (5.5 - 5)^2} = 0.73$$

$$(4.5,5) \text{ vs } (4.125,5.5): \sqrt{(4.125 - 4.5)^2 + (5.5 - 5)^2} = 0.54$$

$$(3.5,4.5) \text{ vs } (4.125,5.5): \sqrt{(4.125 - 3.5)^2 + (5.5 - 4.5)^2} = 1.08$$

step – 3: Form the table

Data point	Distance values with Centroid 1	Distance values with Centroid 2
(1,1)	1.57	7.21
(1.5,2)	0.47	6.1
(3,4)	2.04	1.78
(5,7)	5.64	1.84
(3.5,5)	3.15	0.73
(4,5,5)	3.78	0.54
(3.5,4.5)	2.74	1.08

Data point	Distance values with Centroid 1	Distance values with Centroid 2	Belongs to which group
(1,1)	1.57	7.21	Group-1
(1.5,2)	0.47	6.1	Group-1
(3,4)	2.04	1.78	Group-2
(5,7)	5.64	1.84	Group-2
(3.5,5)	3.15	0.73	Group-2
(4,5,5)	3.78	0.54	Group-2
(3.5,4.5)	2.74	1.08	Group-2

Step – 4: re arrange the group

Group – 1 : (id: 1, id: 2) observations

(1,1) (1.5,2)

Group – 2 : (id: 3, id: 4, id: 5, id: 6, id: 7) observations

(3,4)(5,7) (3.5,5) (4.5,5), (3.5,4.5)

Step – 5: Repeat the 1,2,3,4 steps

$$\text{Mean}(G_1) = \frac{(1 + 1.5)}{2}, \frac{1 + 2}{2} = (1.25, 1.5)$$

$$\text{Mean}(G_2) = \frac{(3 + 5 + 3.5 + 4.5 + 3.5)}{5}, \frac{4 + 7 + 5 + 5 + 4.5}{5} = (3.9, 5.1)$$

Data point	Centroid 1 1.25,1.5	Centroid 2: 3.9,5.1	Belongs to which group
(1,1)	0.56	7.21	Group-1
(1.5,2)	0.56	6.1	Group-1
(3,4)	3.05	1.42	Group-2
(5,7)	6.66	2.20	Group-2
(3.5,5)	4.16	0.42	Group-2
(4,5)	4.78	0.61	Group-2
(3.5,4.5)	3.75	0.72	Group-2

Conclusion : after multiple iterations

Mean values are fixed (1.25,1.5),(3.9,5.1)

Final two clusters

Group – 1 : (1,2)observations

(1,1) (1.5,2)

Group – 2 : (3,4,5,6,7)observations

(3,4)(5,7) (3.5,5) (4,5), (3.5,4.5)